# OptiMOS™3 Power-Transistor

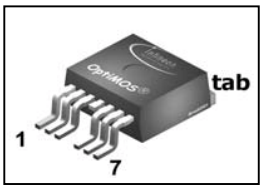
## Features

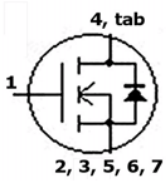
- MOSFET for ORing and Uninterruptible Power Supply
- Qualified according to JEDEC[1)] for target applications
- N-channel
- Normal level
- Ultra-low on-resistance $R_{DS(on)}$
- Avalanche rated
- Pb-free plating; RoHS compliant
- Halogen-free according to IEC61249-2-21

## Product Summary

| | | |
|---|---|---|
| $V_{DS}$ | 40 | V |
| $R_{DS(on),max}$ | 1.1 | mΩ |
| $I_D$ | 180 | A |

PG-TO263-7




| Type | Package | Marking |
|---|---|---|
| IPB011N04N G | PG-TO263-7 | 011N04N |

**Maximum ratings,** at $T_j$=25 °C, unless otherwise specified

| Parameter | Symbol | Conditions | Value | Unit |
|---|---|---|---|---|
| Continuous drain current | $I_D$ | $V_{GS}$=10 V, $T_C$=25 °C | 180 | A |
| | | $V_{GS}$=10 V, $T_C$=100 °C | 180 | |
| Pulsed drain current[2)] | $I_{D,pulse}$ | $T_C$=25 °C | 1260 | |
| Avalanche current, single pulse[3)] | $I_{AS}$ | $T_C$=25 °C | 180 | |
| Avalanche energy, single pulse | $E_{AS}$ | $I_D$=180 A, $R_{GS}$=25 Ω | 610 | mJ |
| Gate source voltage | $V_{GS}$ | | ±20 | V |

[1)] J-STD20 and JESD22

**Maximum ratings,** at $T_j$=25 °C, unless otherwise specified

| Parameter | Symbol | Conditions | Value | Unit |
|---|---|---|---|---|
| Power dissipation | $P_{tot}$ | $T_C$=25 °C | 250 | W |
| Operating and storage temperature | $T_j$, $T_{stg}$ | | -55 ... 175 | °C |
| IEC climatic category; DIN IEC 68-1 | | | 55/175/56 | |

| Parameter | Symbol | Conditions | Values | | | Unit |
|---|---|---|---|---|---|---|
| | | | min. | typ. | max. | |
| **Thermal characteristics** | | | | | | |
| Thermal resistance, junction - case | $R_{thJC}$ | | - | - | 0.6 | K/W |
| SMD version, device on PCB | $R_{thJA}$ | minimal footprint | - | - | 62 | |
| | | 6 cm² cooling area[4)] | - | - | 40 | |
| **Electrical characteristics,** at $T_j$=25 °C, unless otherwise specified | | | | | | |
| **Static characteristics** | | | | | | |
| Drain-source breakdown voltage | $V_{(BR)DSS}$ | $V_{GS}$=0 V, $I_D$=1 mA | 40 | - | - | V |
| Gate threshold voltage | $V_{GS(th)}$ | $V_{DS}$=$V_{GS}$, $I_D$=200 µA | 2 | - | 4 | |
| Zero gate voltage drain current | $I_{DSS}$ | $V_{DS}$=40 V, $V_{GS}$=0 V, $T_j$=25 °C | - | - | 2 | µA |
| | | $V_{DS}$=40 V, $V_{GS}$=0 V, $T_j$=125 °C | - | - | 200 | |
| Gate-source leakage current | $I_{GSS}$ | $V_{GS}$=20 V, $V_{DS}$=0 V | - | - | 200 | nA |
| Drain-source on-state resistance[5)] | $R_{DS(on)}$ | $V_{GS}$=10 V, $I_D$=100 A | - | 0.9 | 1.1 | mΩ |
| Gate resistance | $R_G$ | | - | 1.5 | - | Ω |
| Transconductance | $g_{fs}$ | $\|V_{DS}\|>2\|I_D\|R_{DS(on)max}$, $I_D$=80 A | 110 | 220 | - | S |

[2)] See figure 3 for more detailed information

[3)] See figure 13 for more detailed information

[4)] Device on 40 mm x 40 mm x 1.5 mm epoxy PCB FR4 with 6 cm2 (one layer, 70 µm thick) copper area for drain connection. PCB is vertical in still air.

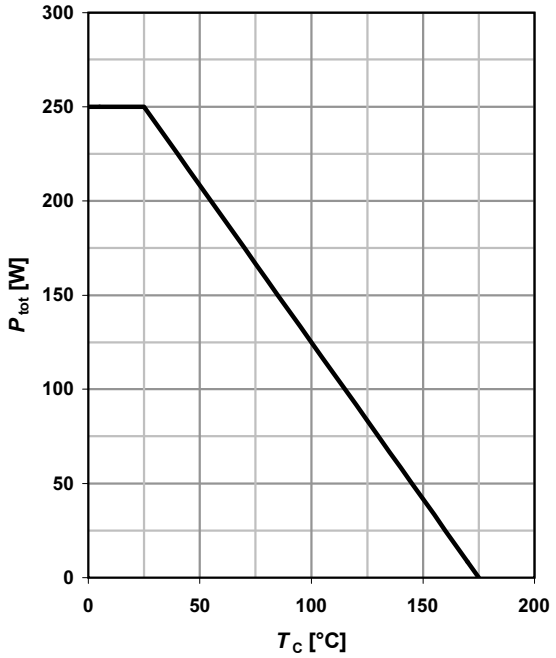
[5)] Measured from drain tab to source pin

| Parameter | Symbol | Conditions | Values | | | Unit |
|---|---|---|---|---|---|---|
| | | | min. | typ. | max. | |
| **Dynamic characteristics** | | | | | | |
| Input capacitance | $C_{iss}$ | $V_{GS}$=0 V, $V_{DS}$=20 V, $f$=1 MHz | - | 16000 | 21000 | pF |
| Output capacitance | $C_{oss}$ | | - | 4000 | 5300 | |
| Reverse transfer capacitance | $C_{rss}$ | | - | 160 | - | |
| Turn-on delay time | $t_{d(on)}$ | $V_{DD}$=20 V, $V_{GS}$=10 V, $I_D$=30 A, $R_G$=1.6 Ω | - | 40 | - | ns |
| Rise time | $t_r$ | | - | 10 | - | |
| Turn-off delay time | $t_{d(off)}$ | | - | 63 | - | |
| Fall time | $t_f$ | | - | 13 | - | |
| Gate Charge Characteristics[5)] | | | | | | |
| Gate to source charge | $Q_{gs}$ | $V_{DD}$=20 V, $I_D$=80 A, $V_{GS}$=0 to 10 V | - | 75 | - | nC |
| Gate charge at threshold | $Q_{g(th)}$ | | - | 46 | - | |
| Gate to drain charge | $Q_{gd}$ | | - | 23 | - | |
| Switching charge | $Q_{sw}$ | | - | 52 | - | |
| Gate charge total | $Q_g$ | | - | 188 | 250 | |
| Gate plateau voltage | $V_{plateau}$ | | - | 4.9 | - | V |
| Gate charge total, sync. FET | $Q_{g(sync)}$ | $V_{DS}$=0.1 V, $V_{GS}$=0 to 10 V | - | 177 | - | nC |
| Output charge | $Q_{oss}$ | $V_{DD}$=20 V, $V_{GS}$=0 V | - | 147 | - | |
| **Reverse Diode** | | | | | | |
| Diode continuous forward current | $I_S$ | $T_C$=25 °C | - | - | 180 | A |
| Diode pulse current | $I_{S,pulse}$ | | - | - | 1260 | |
| Diode forward voltage | $V_{SD}$ | $V_{GS}$=0 V, $I_F$=100 A, $T_j$=25 °C | - | 0.86 | 1.2 | V |
| Reverse recovery charge | $Q_{rr}$ | $V_R$=20 V, $I_F$=$I_S$, $di_F/dt$=400 A/µs | - | - | 180 | nC |

[6)] See figure 16 for gate charge parameter definition

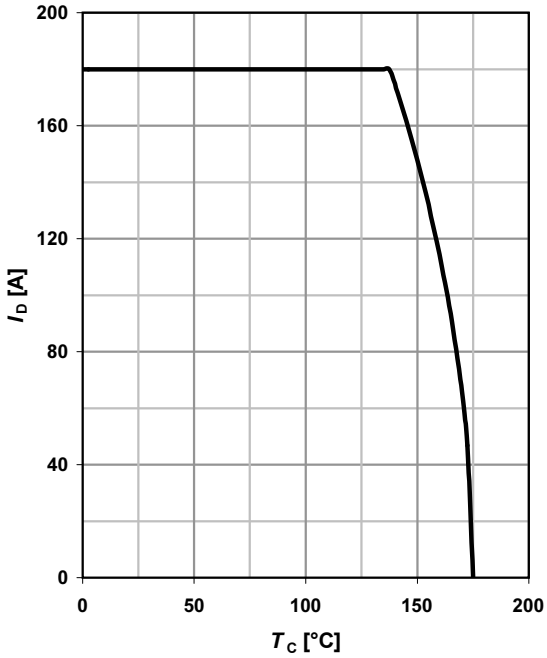
**1 Power dissipation**

$P_{tot}$=f($T_C$)




**2 Drain current**

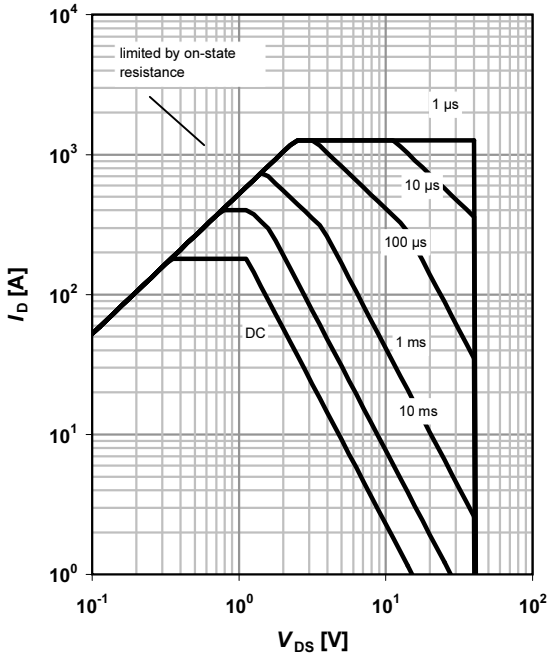
$I_D$=f($T_C$); $V_{GS}$≥10 V




**3 Safe operating area**

$I_D$=f($V_{DS}$); $T_C$=25 °C; $D$=0

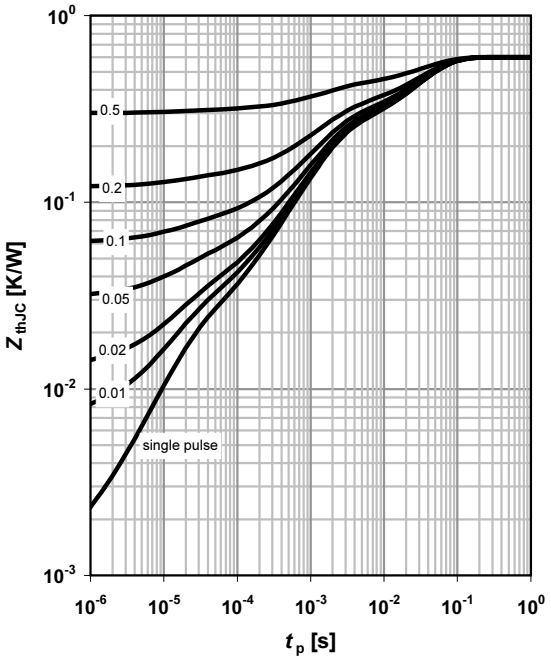
parameter: $t_p$




**4 Max. transient thermal impedance**

$Z_{thJC}$=f($t_p$)

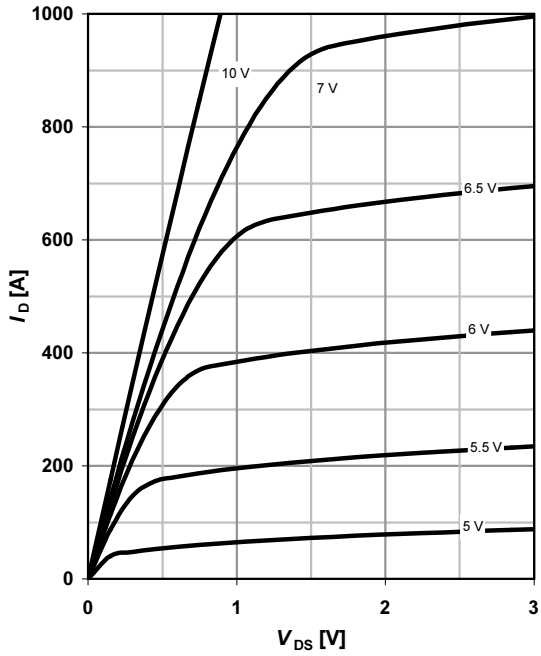
parameter: $D$=$t_p$/$T$



**5 Typ. output characteristics**

$I_D$=f($V_{DS}$); $T_j$=25 °C

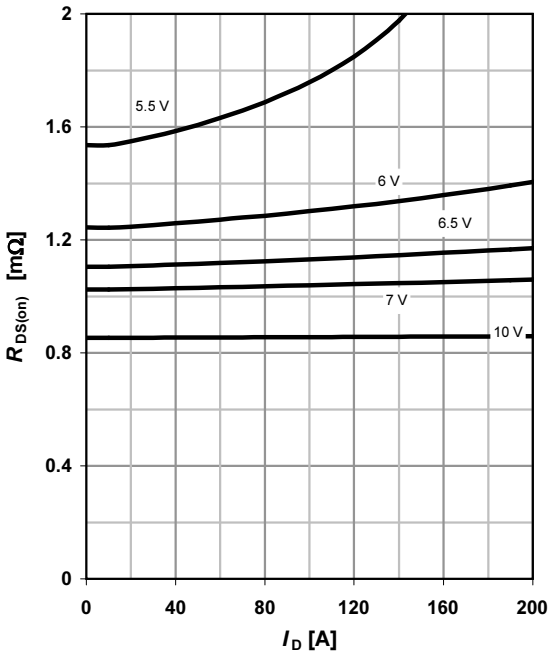
parameter: $V_{GS}$




**6 Typ. drain-source on resistance**

$R_{DS(on)}$=f($I_D$); $T_j$=25 °C

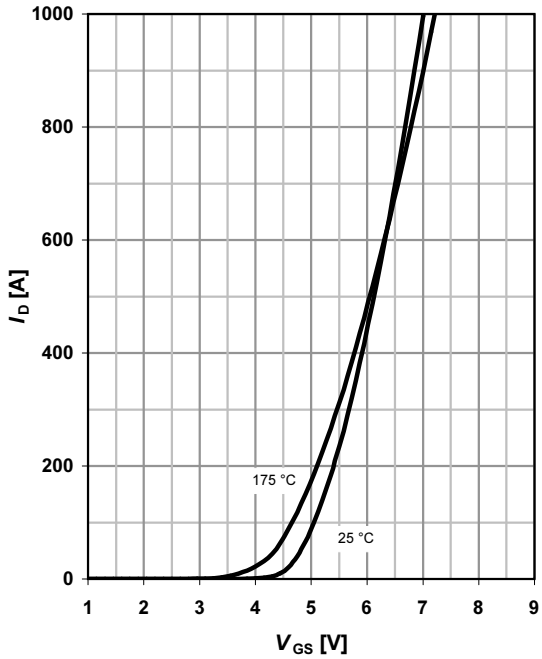
parameter: $V_{GS}$




**7 Typ. transfer characteristics**

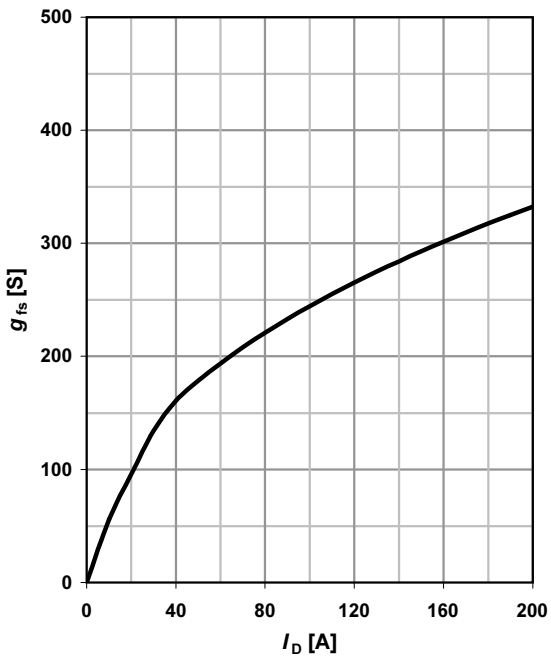
$I_D$=f($V_{GS}$); |$V_{DS}$|>2|$I_D$|$R_{DS(on)max}$

parameter: $T_j$



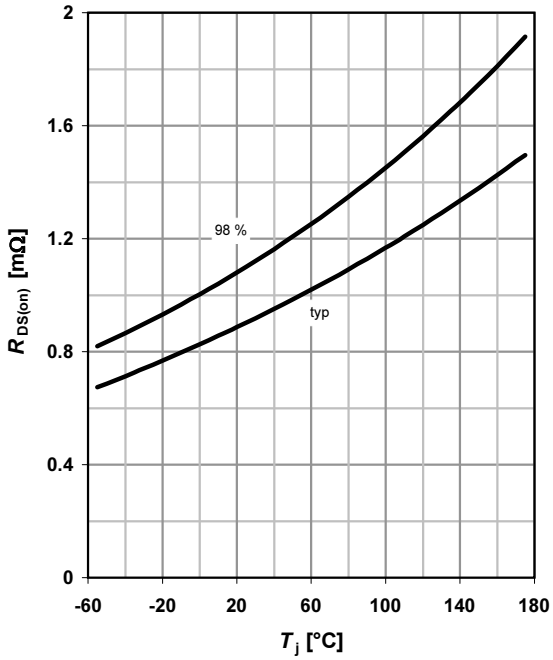

**8 Typ. forward transconductance**

$g_{fs}$=f($I_D$); $T_j$=25 °C



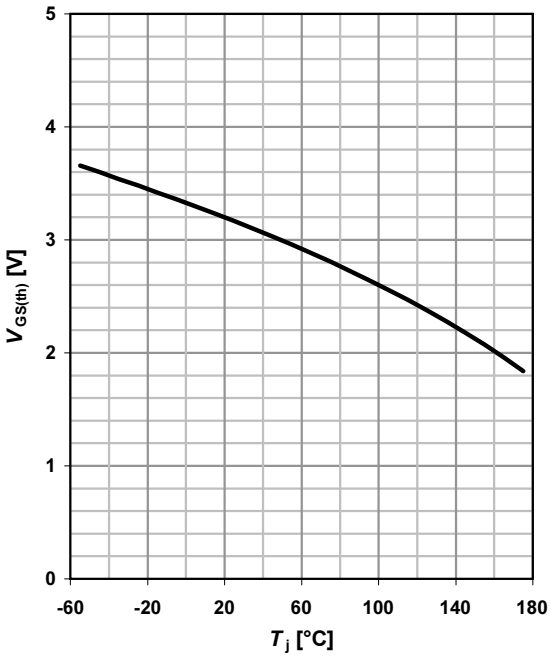
**9 Drain-source on-state resistance**

$R_{DS(on)}$=f($T_j$); $I_D$=100 A; $V_{GS}$=10 V



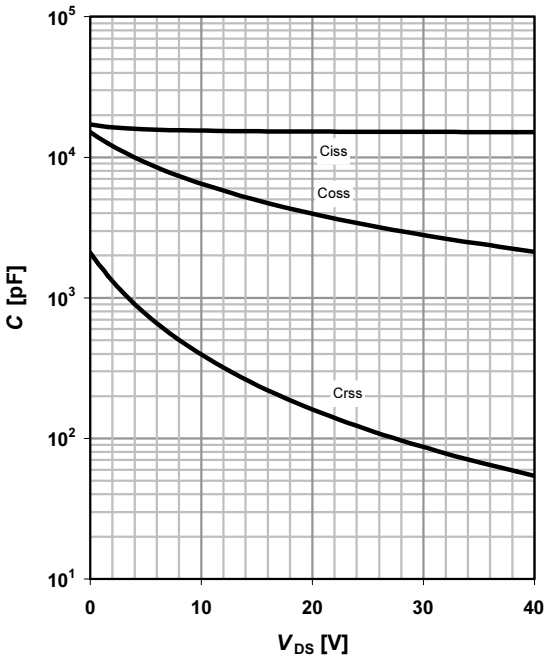

**10 Typ. gate threshold voltage**

$V_{GS(th)}$=f($T_j$); $V_{GS}$=$V_{DS}$; $I_D$=1mA




**11 Typ. capacitances**

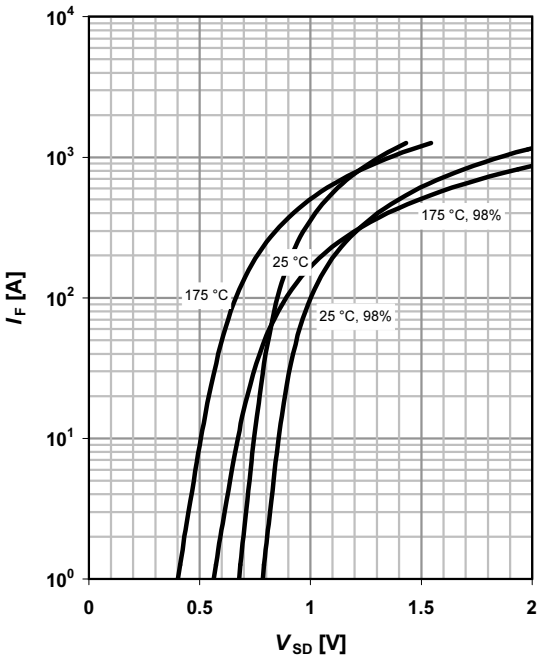
$C$=f($V_{DS}$); $V_{GS}$=0 V; $f$=1 MHz




**12 Forward characteristics of reverse diode**

$I_F$=f($V_{SD}$)

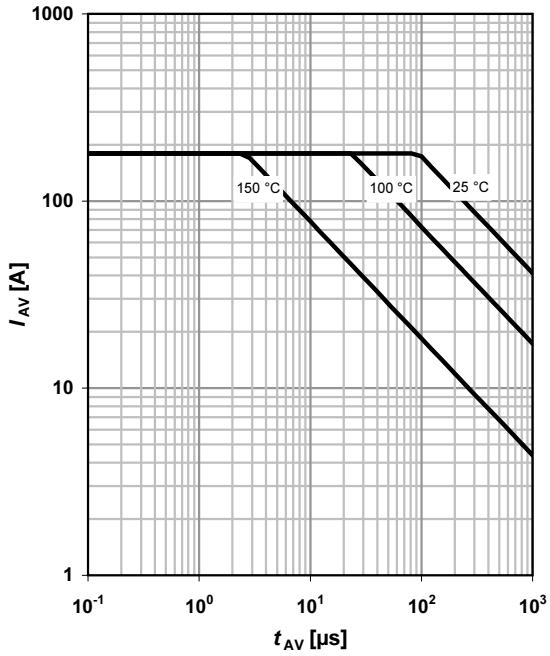
parameter: $T_j$



**13 Avalanche characteristics**

$I_{AS}$=f($t_{AV}$); $R_{GS}$=25 Ω

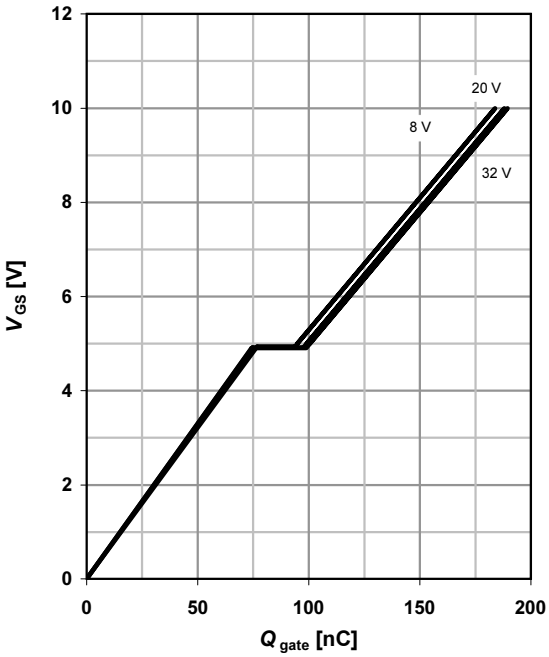
parameter: $T_{j(start)}$




**14 Typ. gate charge**

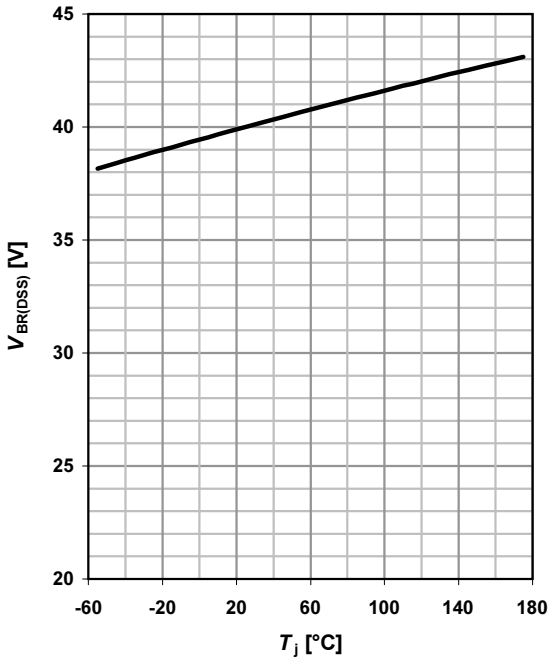
$V_{GS}$=f($Q_{gate}$); $I_D$=80 A pulsed

parameter: $V_{DD}$

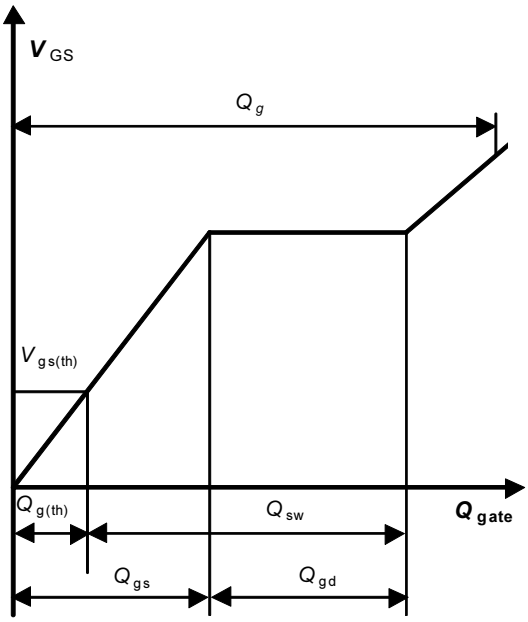



**15 Drain-source breakdown voltage**

$V_{BR(DSS)}$=f($T_j$); $I_D$=1 mA

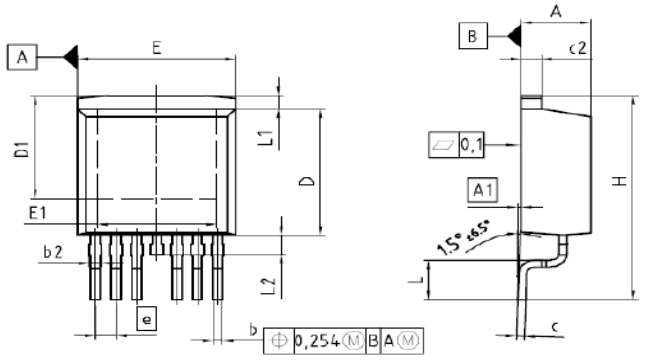



**16 Gate charge waveforms**

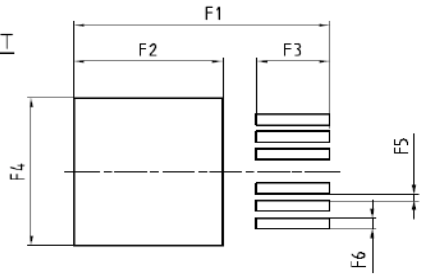


**Package Outline** **PG-TO263-7**

TO263-7-2 / TO263-7-3 / TO263-7-5 / TO263-7-7




FOOTPRINT




| DIM | MILLIMETERS | | INCHES | |
|---|---|---|---|---|
| | MIN | MAX | MIN | MAX |
| A | 4.30 | 4.57 | 0.169 | 0.180 |
| A1 | 0.00 | 0.25 | 0.000 | 0.010 |
| b | 0.50 | 0.70 | 0.020 | 0.028 |
| b2 | 0.50 | 1.00 | 0.020 | 0.039 |
| c | 0.33 | 0.65 | 0.013 | 0.026 |
| c2 | 1.17 | 1.40 | 0.046 | 0.055 |
| D | 8.51 | 9.45 | 0.335 | 0.372 |
| D1 | 6.90 | 7.90 | 0.272 | 0.311 |
| E | 9.80 | 10.31 | 0.386 | 0.406 |
| E1 | 6.50 | 8.60 | 0.256 | 0.339 |
| e | 1.27 | | 0.050 | |
| N | 6 | | 6 | |
| H | 14.61 | 15.88 | 0.575 | 0.625 |
| L | 2.29 | 3.00 | 0.090 | 0.118 |
| L1 | 0.70 | 1.60 | 0.028 | 0.063 |
| L2 | 1.00 | 1.78 | 0.039 | 0.070 |
| F1 | 16.05 | 16.25 | 0.632 | 0.640 |
| F2 | 9.30 | 9.50 | 0.366 | 0.374 |
| F3 | 4.50 | 4.70 | 0.177 | 0.185 |
| F4 | 10.70 | 10.90 | 0.421 | 0.429 |
| F5 | 0.37 | 0.57 | 0.015 | 0.022 |
| F6 | 0.70 | 0.90 | 0.028 | 0.035 |

DOCUMENT NO.
Z8B00134765

SCALE 0 5 7.5mm

EUROPEAN PROJECTION

ISSUE DATE
05-11-2007

REVISION
01

**Published by**
**Infineon Technologies AG**
**81726 Munich, Germany**
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